Introduction to Internet of Things(IoT)

Infomax
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Introduction

- IoT stands for Internet of things which means accessing and controlling daily usable equipment and devices using internet.
- It is a platform where everyday devices become smarter, everyday processing become intelligent, and everyday communication become informative.
- IoT is a system of interconnected objects, usually called smart devices, through the internet.
- Before the invention of IoT, there were only two type of communication either human-to-human or human-to-devices, but the invention of IoT made it possible to establish communication among Machine-to-machine(M2M) without any human interaction.
- IoT was first developed in year *1999* by *Kevin Ashton*, one of the founders of the Auto-ID Center as MIT.

Advantages of IOT Applications

- Access information
- Communications
- Cost-effective and efficient
- Automation
- Minimize human effort
- Save time
- Useful in monitoring

Disadvantages of IoT Applications

- Security
- Complexity
- Unemployment
- Dependability
- Becoming Lazy

Characteristics of IoT

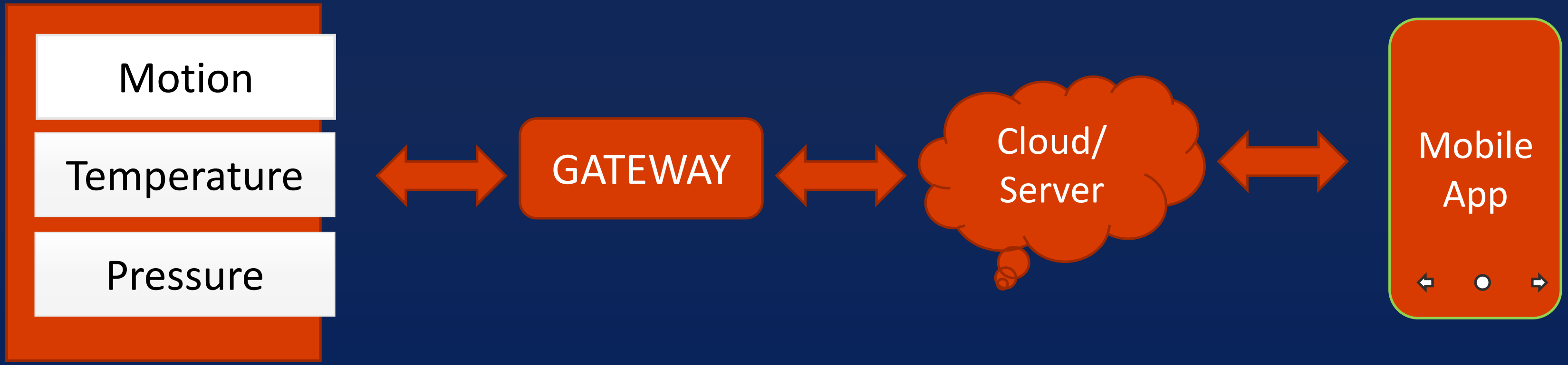
- Connectivity
- Things-related services
- Heterogeneity
- Interoperability
- Dynamic Changes
- Enormous scale
- Safety

Application of IoT

- Home Automation
- Smart water System
- Smart Transport
- Smart Cities
- Social life and Entertainment
- Health and Fitness
- Smart Environment and Agriculture
- Livestock Monitoring
- Smart Greenhouse

Working and Implementation of IoT

➢ The IoT basically consists of web-enabled devices that can read or collect, send and respond to the data they acquired from the environment by the use of highly functioning, sensors, hardware that can communicate, network connection and processor.



Components of an IoT System

- IoT component include hardware such as the sensor, communication hardware etc. Middleware such as data analytical tools and storage and presentation for delivery of information to the end user.
- There are four fundamental component of an IoT system:
  - Sensors/Devices
  - Connectivity
  - Data Processing
  - Delivery of information

Sensor/Devices

- Sensor are devices are a key component that helps you to collect live data from the surrounding environment.
- Type of Sensors:
  - Temperature sensor
  - Motion sensor
  - Air quality sensor
  - Moisture sensor
  - Light sensor

Connectivity

- Connectivity element plays a very significant task in IoT.
- The communication enables messaging for the devices.
- All the collected data is sent to a cloud infrastructure where devices connect to the cloud to perform different operations.
- The sensor can be connected to the cloud through various mediums of communication and transport such as cellular networks, satellite network, Wi-Fi, Bluetooth, wide are networks(WAN), Low Power wide area network etc. and the information that is sensed at machine level is further transmitted to the cloud-based service for sequential processing.

Data Processing

- Once the data is collected, and it gets to the cloud, the software performs processing on the gathered data.
- This process can be checking the temperature, reading on devices like AC or heaters etc.

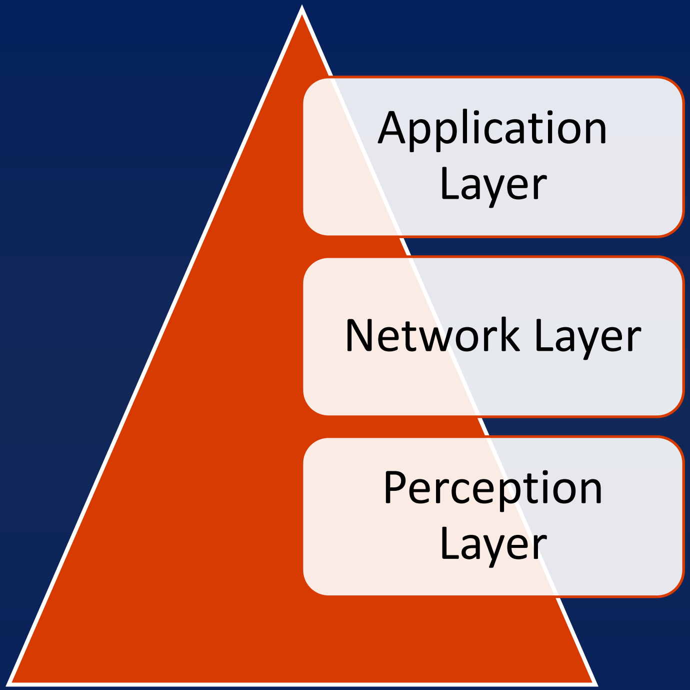
Delivery of information

- Delivery of information is the last element of IoT where the data after being processed ate delivered to the end-user in some way which can be achieved by triggering alarms on their phone or sending them notification through email or text message.

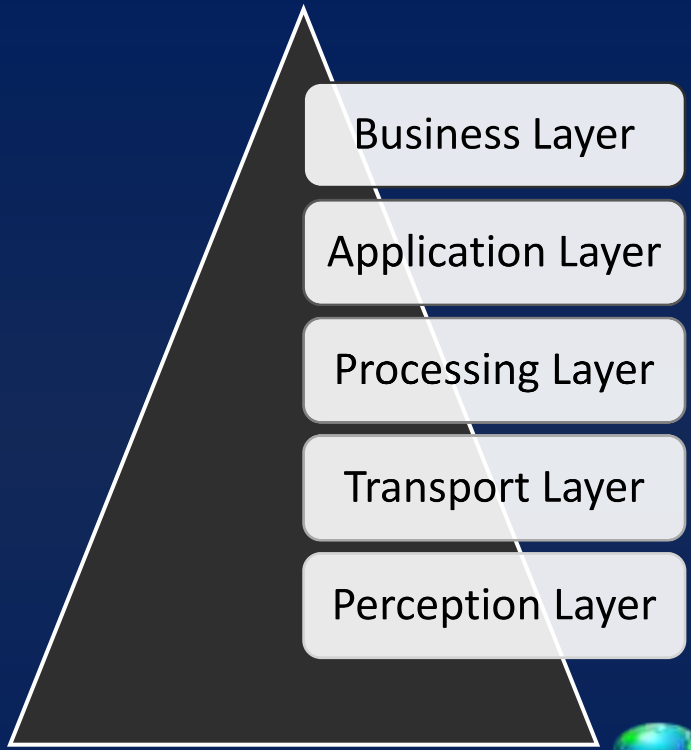
IoT architecture and levels

- IoT architecture shows the building blocks of an IoT system and how they are connected to collect, store and process data.
- IoT architecture also contains device and user management components to provide stable and secure functioning of things and control user access issues.
- The most basic architecture is a three-layer architecture which introduced in the early stages of research in this area.
- But it is not sufficient for research on IoT because research often focuses on finer aspects of the internet of things.

Three and Five-Layer Architectures



Three Layer IoT Architecture



Five Layer IoT Architecture

Perception(recognition) Layer

- This layer is the physical layer, which has sensor for sensing and gathering information about the environment.
- One of the important function of this layer is that it linked and converts the information gathered by the sensors into digital signals which are further transferred to the Network layer.
- Example : The smart phone itself has many type of sensors embedded in it such as the location sensor(GPS), movement sensor(accelerometer, gyroscope), camera, light sensor, microphone, proximity sensor, and magnetometer.

Network Layer

- The Network layer is responsible for connecting to other smart things, network devices, and servers.
- The function of this layer are to receive and transmit the digital data to the processing (middleware) layer.
- These data are transmitted onto the middleware layer over some transmission medium such as WI-FI,4G and GSM etc along with protocol like IPv4,DDS etc.

Processing Layer

- The data that are received from the network layer is processed in this layer.
- This layer uses the technologies like cloud computing, big data processing etc.
- The processing layer is also known as the *Middleware Layer*.

Application Layer

- This layer is responsible for delivering application specific services to the user.
- It define various application in which the IoT can be deployed, based on the processed data.
- Example:
  - Smart phone, smart watch etc.

Business Layer

- This layer acts as the manager for the application and services of the IoT

IoT Ecosystem

- The IoT ecosystem is a connection of various kind of devices that sense and analyze the data and communicates with each over the networks.
- In the IoT ecosystem, the use uses smart devices such as Smartphone, tablet, sensors, etc. to send the command or request to devices over the networks.

Component of ecosystem:

The IoT ecosystem include all the component that enable businesses, governments, and consumer to connect to their IoT devices.

Components of IoT Ecosystem

- Gateway
- Analytics
- Connectivity of devices
- Cloud
- User interface
- Dashboard
- Standard and protocols
- Database
- Automation
- Development

Type of Network

- Body Area Network (BAN)
- Personal Area Network (PAN)
- Home Area Network (HAN)
- Neighbourhood Area Network (NAN)
- Local Area Network (LAN)
- Metropolitan Area Network (MAN)
- Wide Area Network (WAN)
- Virtual Private Network (VPN)
- Campus Area Network(CAN)
- Global Area Network (GAN)

Technologies used in IoT

- Bluetooth
- Bluetooth low Energy (BLE)
- Wi-Fi (wireless Fidelity)
- Low Power Wi-Fi (WiFi-Halow)
- LiFi (Light Fiselity)
- Cellular Networks
- Z-Wave
- RFID (Radio Frequency Identification)
- X-10

Technologies used in IoT (Cont.)

- Sigfox
- BlZigbee
- Low range Wide Area Network (LoRaWAN)
- 6LoWpan
- 5G
- Low Pawer Wide Area Networks (LPWAN)
- Thread
- Near Field Communication (NFC)
- (Global System for Mobile Communication (GSM)

Technologies used in IoT (Cont.)

- General Packet Radio Service (GPRS)
- Long-term Evolution Advanced (LTE-A)
- Wireless Sensor Network (WSN)

Bluetooth

- Bluetooth is a wireless personal Area Network (WPAN) protocol designed by the Bluetooth Special interest Group (SIG).
- It is an open standard for short-range transmission of digital voice and data.
- It operates in the frequency band of 2.45 GHz.
- Data transfer rate of Bluetooth is 2.1 Mbps.

Bluetooth low Energy (BLE)

- The Bluetooth low Energy is commonly known as BLE or Bluetooth 4.0 and marketed as Bluetooth Smart.
- It is a wireless standard and is intended to exchange data over short distance and build Personal Area Networks(PANs).
- Due to improved energy consumption, it is suited for sensor and other devices that require low power.
- The devices usually transmit a small amount of data as bursts at 1Mbps speed.
- It is used in application such as smart building, smart transportation, Smart phones, fitness trackers and wearable.

Wi-Fi (Wireless Fidelity)

- Wi-Fi is wireless technology that allows an electronic devices to exchange data wirelessly (using radio waves) over a computer network, including high-speed internet connections.
- Wi-Fi technology enables IoT wireless connectivity over a LAN using IEEE 802.11 standards.
- Wi-Fi standard 802.11 offers transfer rate of 100 Mbps, making it ideal for data transfer but too power consuming for some IoT applications.
- Wi-Fi is primarily a local area networking (LAN) technology.
- The frequency of Wi-Fi is between 2.4 GHz to 5 GHz.

Low Power Wi-Fi (WiFi-HaLow)

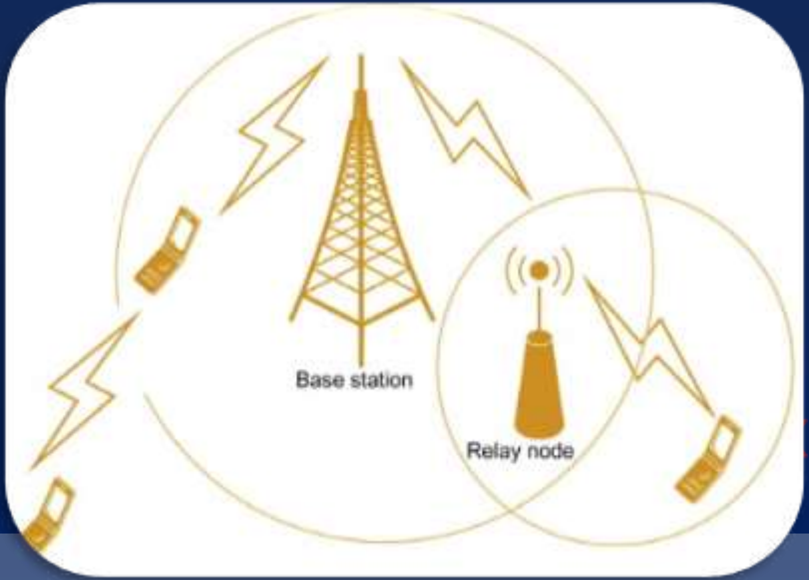
- The original Wi-Fi standard are not suitable for IoT application sue to their frame overhead and high power consumption.
- Hence, IEEE 802.11 working group initiated 802.11ah task group to develop a standard that supports low overhead, power friendly communication suitable for sensor and remotes.
- The WiFi alliance has recently developed 'WiFi HaLow' which is based on the IEEE 802.11ah standard.
- The range of WiFi HaLow is nearly twice that of traditional WiFi.
- Designed specifically for low data rate, long range sensors and controllers.

LiFi (Light Fidelity)

- It is new wireless technology that offers high speed data transmission based on Visible Light communication (VLC) principal.
- It uses two main component LiFi controller or coordinator and LiFi dongles.
- The technology uses LED lamps on source side and photo detector based LiFi dongles on destination side.
- Li-Fi can also utilize energy efficent Light-Emitting Diode (LED) light to reduce energy cost.
- Lifi uses light as medium for data communication.
- The speed of data provided by by these LEDs is 100 times faster than the WiFi speed.

Cellular Network

- A cellular network is a radio network distributed over land through cells where each includes a fixed location transceiver known as base station.
- Cellular technology is the basis of mobile phone networks.
- These network exists in various generations such as 3G,4G and 5G of cellular network.



Z-Wave

- Z-Wave is a wireless network standard that consumes very low power and it is widely used for connecting smart devices in the IoT like home automation, lighting controls, security systems, Energy saving etc.
- Uses low-energy radio waves to communicate.
- Applicable for less data speed and small data size applications.
- Cannot be deployed over a wide area as the range is less.

RFID (Radio Frequency Identification)

- RFID is small electronic devices that consist of small chip and antenna.
- RFID tags store identifiers and data.
- RDIF is used in indoor moving navigation, smart parking, supply chain management, and battery-less remote control devices applications.

X-10

- X-10 is a protocol for communication among IoT devices, mainly in the home automation sector.
- It is called as the father of modern home automation.
- X-10 provide a remote interface so that lights and other appliances can be turned *on* and *off*.

Sigfox

- Sigfox is a cellular style system that enables remote devices to connect using ultra-narrow band, to provide low power low data rate, and low cost communications for remote connected devices.
- Sigfox uses a technology called Ulta Narrow Band (UNB) .
- Sigfox designed to handle low data-transfer speeds of 10 to 1,000 bits per second.
- Sigfox is idle for IoT applications that need to send infrequent and small bursts of data, such as alarm systems, smart meters, smart street lighting, patient monitors etc.

Zigbee

- Zigbee is a low powered, low cost standard on a wireless mesh network designed for battery powered devices and cheaper alternative to bluetooth.
- It is based on the IEEE 802.15.4 communication protocol standard and used for personal area networks(PAN).
- The range of Zigbee device communication is very small (10-100 meters).

Low Range Wide Area Network (LoRaWAN)

- LoRaWAN is newly developed long range wide area network wireless technology designed for IoT application with power saving, low cost, mobility, security, and bi-directional communication requirements.
- LoRaWAN defines the communication protocol and system architecture for the network in the Medium Access Control (MAC) layer protocol.
- It is ideal for smart cities and industrial applications.
- It is not ideal to be used for real time applications.

6LoWpan

- 6LoWpan is acronym of IPv6 over Low Power Wireless Personal Area Network.
- It is a protocol that is primarily intended for log-range wireless battery-operated IoT devices in regional, national, or global networks.
- The 6LoWpan standard enables any low power radio to communicate to the internet, including 804.15.4, Bluetooth Low energy and z-wave for home automation applications.

5G

- This is 5th generation of wireless, popularly known as 5G. The “G” stands for generation of mobile technology.
- 5G is a medium range communication that offers many improvement over 4G(called LTE-advanced), such as increased speed and better coverage.
- It provides high data transfer rate and more effective and efficient.

Low Power Wide Area Networks (LPWAN)

- It is a type of wireless network used for low power IoT and M2M applications.
- It defines low power area networks (LPWAN) standard to enable IoT and can transmit data over long distance while having low power consumption .
- The LPWAN network offers longer range from 5 Km to 30 Km.

Thread

- A very new IP-based IPv6 networking protocol aimed at the home automation environment is Thread.
- Thread used IPv6 for the Internet protocol addressing and each of the devices is IP-addressable.
- Thread enables device-to-device and device-to-cloud communications and reliably connects thousands of products and include mandatory security features.
- Thread is based on the broadly supported IEEE 802.15.4 radio standard, which is designed from the ground up for extremely low power consumption and latency.

Near Field Communication (NFC)

- NFC is a set of the communication-based protocol used for communication between two devices in the range of approximately 10 cm.
- All type of data can be transferred between two NFC enabled devices in second by bringing them close to each other.
- The most used applications are smart phones, contactless payment system, parking meters, e-ticket booking etc.
- NFC is an expensive technology.

GSM (Global System for Mobile communications)

- GSM is digital cellular phone technology .
- Developed in the 1980s, GSM was first deployed in seven European countries in 1992.
- GSM provides a short messaging service(SMS) that enables text message up to 160 characters in length to be send using GSM phone.
- GSM phone use a Subscriber Identity Module(SIM) smart card that contain user account information.
- GSM is an open, digital cellular technology used for transmitting mobile voice and data service.

GPRS (General Packet Radio Service)

- It is first high-speed digital data service provided by cellular carriers that used the GSM technology.
- GPRS works on GPRS cell phones as well as laptops and portable devices that have GPRS modems.
- User have typically experienced downstream data rates up to 80 Kbps.
- It allows simultaneous use of both voice and data services. Hence, user can have both voice call and data call together.

LTE-A (Long-term Evolution Advanced)

- LTA-A or LTE advanced, delivers an important upgrade to LTE technology by increasing not only its coverage, but also reducing its latency and raising its throughput.
- LTE-A is a collection of cellular networking standards that is designed to meet M2M and IoT requirements in such networks.
- It is one of the most scalable and cost effective standard compared to other cellular protocol.

WSN (Wireless Sensor Network)

- A WSN is the network that are wireless in nature which comprises of distributed devices with sensors which are used to sense or monitor the environmental and physical conditions like temperature, gyroscope, pressure, etc. and make the embedded processing using these devices.
- These devices are connected through different devices such as GPS, WiFi, RFID, etc. over the networks.
- The most used application of WSN are weather monitoring system, Soil moisture monitoring system, Surveillance system, health care monitoring, Indore air quality monitoring system etc.

Communication Protocols

- Communication protocols form the backbone of IoT system and network connectivity.
- Communication protocol allows devices to exchange data over the network.
- Protocol are usually classified according to the layer they correspond to in the Open System interconnection (OSI) reference model for networking. Type of protocols include the following :
  - Data-link protocols
  - Communication or Network protocols
  - Transport layer protocols
  - Application layer protocols

Application Layer Protocols

- CoAP(Constrained Application Protocol) : CoAP is an internet-utility protocol created specifically for connecting devices with limited (constrained) resources such as a small memory or short battery life.
- MQTT (message Queue Telemetry Transport) : MQTT is an open light-weight easy-to-implement messaging protocol for M2M communication.
- XMPP (Extensible Messaging and Presence Protocol) : XMMP is a open technology for real-time communication original designed for chats and message exchange applications.
- DDS (Data Distribution Service): DDS is an IoT standard for read-time, scalable and high-performance machine-to-machine communication.

Application Layer Protocols

- AMQP (Advanced Message Queuing Protocol) : AMQP is an open source published standard for asynchronous messaging by write.
- Web Socket : Web Socket is a protocol that provide bi-directional, full-duplex communication channels, over a single Transmission Control Protocol (TCP) sockets.
- HTTP (Hyper Text Transfer protocol) : HTTP is the foundation of the client-server model used for data communication over the web.

Transport Layer Protocols

- TCP(Transmission Control Protocol) : TCP is a standard that defies how to establish and maintain a network conversation via which application program can exchange data. TCP works with the Internet Protocol (IP), which define how computers send packets of data to each other.
- UDP (User Datagram Protocol) : UDP is an alternative communication protocol to TCP used for establishing low-latency and loss-latency and loss tolerating connection between application on the internet. UDP is also better suited for real time data application such as voice and video. UDP is connectionless protocol that works at the transport layer.

Network Layer Protocol

- **IPv4 :** IP (Internet Protocol) is the main networking protocol. Ther are two version of IP(IPv4 and IPv6). IPv4 is limited to 32-bit address which only provide around 3.4 billion address in total, which is less than current number of IoT devices that are connected.
- **IPv6 :** IPv6 is the latest revision of the Internet Protocol (IP), the communications protocol that provides and identification and location system for computers on networks and routes traffic across the internet. IPv6 used 128-bit address.
- **6LoWpan :** 6LoWpan a very popular standard for wireless communication.6LoWpan is a mesh network that is robust and scalable.

Data-link layer Protocols

- 802.3 Ethernet : Ethernet is based upon the IEEE 802.3 standard and is widely used to build LANs.
- 802.11 WiFi : IEEE 802.11 is a collection of wireless Local Area Network (WLAN) communication standard.
- 802.16 WiMAX : IEEE 802.16 is a collection of wireless broadband standards. WiMAX (Worldwide Interoperability for Microwave Access) standards provides data rate from 1.5 Mb/s to 1 Gb/s.
- 802.15.4 LR-WPAN: IEEE 802.15.4 is a collection of Low-Rate Wireless Personal Area Networks (LR-WPAN) standards.
- 2G/3G/4G/5G Mobile Communication: These are different generation of mobile communication standards. IoT devices based on these standards can communicate over cellular network.

IoT Enabling Technologies

- IoT enabling technologies includes :
  - Sensors
  - Cloud computing
  - Big data analytics
  - Communication protocols
  - Embedded system.

Sensors

- A sensor is a device that is used to sense a physical condition or event.
- A Sensor works by converting a non-electrical input into a electrical signal that can be sent to an electronic circuit.

Cloud Computing

- Cloud computing or no-demand computing is an Internet-based computing service which involves the use of the network of remote server rather than local area server or personal computer.
- Cloud computing is a transformative computing paradigm that involves delivering applications and services over the internet.

Big Data Analytics

- Big Data Analytics refers to the process of collecting, organizing, analyzing large data sets to discover different patterns and other useful information.
- Example of big data generated by IoT system are Weather Monitoring system, Machine sensor data from industrial system, Health and fitness data, Location and tracking system etc.
- Characteristics of Big Data
  - Volume
  - Variety
  - Velocity
  - Variability

Communication Protocol

- Communication protocol are format descriptions of digital message formats and rules.
- They are required to exchange messages in or between computing system.

Embedded System

- An embedded system is a system that has computer hardware and software embedded to perform specific task.
- They include a processing unit, usually a general-purpose microcontroller(i.e. CPUs with integrated memory or peripheral interfaces), along with on-board peripheral component such as I/O ports, memory and more.
- Embedded system are small computers that are implemented as part if a large system or product and designed to execute a specific sunction or application.

Fog Computing

- Fog computing also known as Edge Computing is a term created by Cisco and it extends the concept of cloud computing to the network edge, making it ideal for IoT and other applications that require real-time interaction.
- Fog computing has many real-word applications.
    - Smart Traffic management
    - Smart cities and grids
    - Industrial IoT

Building block of IoT

Four things form basic building blocks of IoT devices.

- **Sensor & actuators** : Sensor are devices which converts physical parameters like temperature, motion etc. into the electronic signal. Actuators are devices which is a contrast to sensor. It transform electrical signal into physical movement.
- **Processor** : processors are the brain of the IoT system. Their main function is to process the data captured by the sensor.
- **Gateways** : Gateway are responsible for routing the processed data and send it to proper location for its (data) proper utilization.
- **Applications** : Application form another end of an IoT system. Application are essential for proper utilization of all the data collected.

Functional Block of IoT

- An IoT system comprises of a number of functional block that provide the system the capabilities for identification, sensing, actuation. communication, and management.

These functional blocks are as follows:

- Device: an IoT devices provide sensing, actuation, monitoring and control function.
- Communication: It handles communication for the IoT system.
- Security: It secures the IoT system by providing authentication, authorization, message and content integrity, and data security.
- Application : It provides an interface that the user's can use to control and monitor various aspects of IoT system.
- Services: An IoT system uses various type of IoT services such as service for device monitoring ,device control services and data publishing.

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